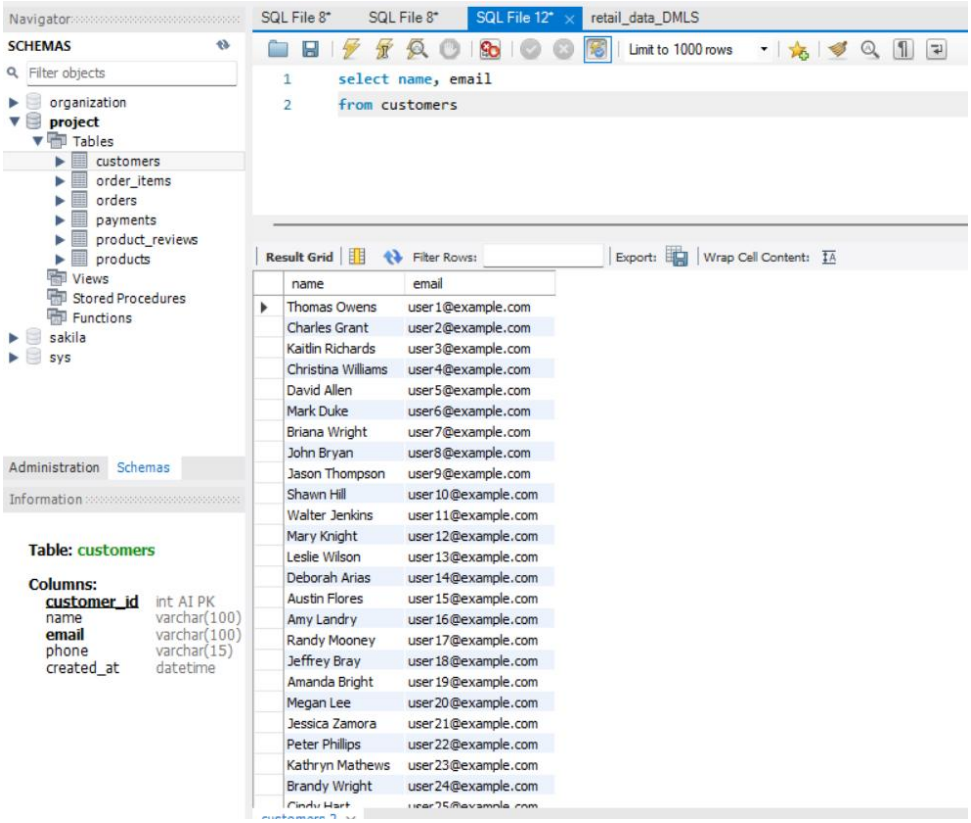
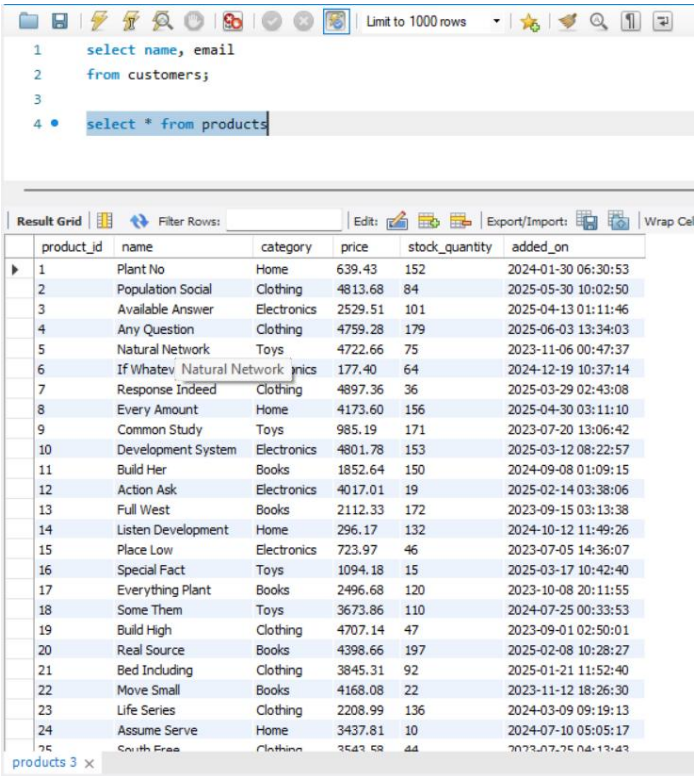
Level 1: Basics

.

1
select name, email
from customers;



2
select * from products;



3
select distinct Category from products;

```
from customers;

select * from products;

select distinct Category from products
```

Result Grid

Category
Home
Clothing
Electronics
Toys
Books

4
select name, price from products
where price > 1000;

```
select * from products;

select distinct Category from products;

select name, price from products
where price > 1000
```

Result Grid

name	price
Population Social	4813.68
Available Answer	2529.51
Any Question	4759.28
Natural Network	4722.66
Response Indeed	4897.36
Every Amount	4173.60
Development System	4801.78
Build Her	1852.64
Action Ask	4017.01
Full West	2112.33
Special Fact	1094.18
Everything Plant	2496.68
Some Them	3673.86
Build High	4707.14
Real Source	4398.66
Bed Including	3845.31
Move Small	4168.08
Life Series	2208.99
Assume Serve	3437.81
South Free	3543.58
Movement Future	3502.62
Answer But	1016.68
East Foot	2414.93
Drop Remember	4160.45

products 5

5
select product_id, name, category,price from products
WHERE price between 2000 AND 5000;

```
select name, price from products
where price > 1000;

select product_id, name, category,price from products
WHERE price between 2000 AND 5000
```

Result Grid

product_id	name	category	price
2	Population Social	Clothing	4813.68
3	Available Answer	Electronics	2529.51
4	Any Question	Clothing	4759.28
5	Natural Network	Toys	4722.66
7	Response Indeed	Clothing	4897.36
8	Every Amount	Home	4173.60
10	Development System	Electronics	4801.78
12	Action Ask	Electronics	4017.01
13	Full West	Books	2112.33
17	Everything Plant	Books	2496.68
18	Some Them	Toys	3673.86
19	Build High	Clothing	4707.14
20	Real Source	Books	4398.66
21	Bed Including	Clothing	3845.31
22	Move Small	Books	4168.08
23	Life Series	Clothing	2208.99
24	Assume Serve	Home	3437.81
25	South Free	Clothing	3543.58
26	Movement Future	Electronics	3502.62
28	East Foot	Toys	2414.93
29	Drop Remember	Electronics	4160.45
30	Future Sort	Home	3043.67
31	Series Page	Electronics	2070.37
32	Factor Where	Clothing	2295.14

products 6

6
select * from customers
where customer_id in (4,6,8);

SQL File 8* | SQL File 8* | SQL File 12* | retail_data_DMLS

```
select product_id, name, category,price from products
WHERE price between 2000 AND 5000;

select * from customers
where customer_id in (4,6,8)
```

Result Grid

customer_id	name	email	phone	created_at
4	Christina Williams	user4@example.com	586-605-5061x06	2024-10-27 17:19:38
6	Mark Duke	user6@example.com	(144)957-2811	2024-06-24 03:22:59
8	John Bryan	user8@example.com	045.568.0798x27	2025-02-15 17:57:04
NULL	NULL	NULL	NULL	NULL

7
select customer_id, name from customers
where name like 'A%';

SQL File 8* | SQL File 8* | SQL File 12* | retail_data_DMLS

Limit to 1000 rows

```
12  WHERE price between 2000 AND 5000;
13
14  select * from customers
15  where customer_id in (4,6,8);
16
17  select customer_id, name from customers
18  where name like 'A%'
```

Result Grid | Filter Rows: | Edit: | Expo

customer_id	name
15	Austin Flores
16	Amy Landry
19	Amanda Bright
27	Adrienne Green
NULL	NULL

Adrienne Green

8
select product_id, name, price from products
where category = 'Electronics' and price < 3000;

SQL File 8* | SQL File 8* | SQL File 12* | retail_data_DMLS

Limit to 1000 rows

```
15  where customer_id in (4,6,8);
16
17  select customer_id, name from customers
18  where name like 'A%';
19
20  select product_id, name, price from products
21  where category = 'Electronics' and price < 3000
```

Result Grid | Filter Rows: | Edit: | E

product_id	name	price
3	Available Answer	2529.51
6	If Whatever	177.40
15	Place Low	723.97
31	Series Page	2070.37
34	Despite Win	1340.34
44	Actually Term	396.11
47	Southern Thing	512.46
NULL	NULL	NULL

9
select name, price from products
order by price desc;

SQL File 8* | SQL File 8* | SQL File 12* | retail_data_DMLS

Limit to 1000 rows

```
17  select customer_id, name from customers
18  where name like 'A%';
19
20  select product_id, name, price from products
21  where category = 'Electronics' and price < 3000;
22
23  select name, price from products
24  order by price desc
```

Result Grid | Filter Rows: | Export: | Wrap Cell C

name	price
Response Indeed	4897.36
Population Social	4813.68
Development System	4801.78
Any Question	4759.28
Fire Often	4734.89
Natural Network	4722.66
Build High	4707.14
Serious Recognize	4523.10
Study Total	4413.68
Real Source	4398.66
Involve Officer	4244.09
Group But	4180.23
Every Amount	4173.60
Move Small	4168.08
Drop Remember	4160.45
Data Add	4063.38
Action Ask	4017.01
Weight Enter	3875.18

10
select name, price from products
order by price desc, name asc;

SQL File 8* | SQL File 8* | SQL File 12* | retail_data_DMLS

Limit to 1000

```
22
23  select name, price from products
24  order by price desc;
25
26  select name, price from products
27  order by price desc, name asc
```

Result Grid | Filter Rows: | Export:

name	price
Response Indeed	4897.36
Population Social	4813.68
Development System	4801.78
Any Question	4759.28
Fire Often	4734.89
Natural Network	4722.66
Build High	4707.14
Serious Recognize	4523.10
Study Total	4413.68
Real Source	4398.66
Involve Officer	4244.09
Group But	4180.23
Every Amount	4173.60
Move Small	4168.08
Drop Remember	4160.45

Level 2: Filtering and Formatting

1

```
select o.order_id, o.customer_id, o.order_date, o.status, o.total_amount from orders o
left join customers c on o.customer_id = c.customer_id
where c.customer_id is null;
```

SQL File 8 | SQL File 8* | SQL File 12* | retail_data_DMLS

Limit to 1000 rows

```
23 •  select name, price from products
24    order by price desc;
25
26 •  select name, price from products
27    order by price desc, name asc;
28
29 •  use project;
30
31 •  select o.order_id, o.customer_id, o.order_date, o.status, o.total_amount from orders o
32    left join customers c on o.customer_id = c.customer_id
33    where c.customer_id is null
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

order_id	customer_id	order_date	status	total_amount

2

```
select name as Customer_Name, email as Customer_Email from customers;
```

SQL File 8 | SQL File 8* | SQL File 12* | retail_data_DMLS

Limit to 1000 rows

```
25
26 •  select name, price from products
27    order by price desc, name asc;
28
29 •  use project;
30
31 •  select o.order_id, o.customer_id, o.order_date, o.status, o.total_amount from orders o
32    left join customers c on o.customer_id = c.customer_id
33    where c.customer_id is null;
34
35 •  select name as Customer_Name, email as Customer_Email from customers
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_Name	Customer_Email
Thomas Owens	user1@example.com
Charles Grant	user2@example.com
Kaitlin Richards	user3@example.com
Christina Williams	user4@example.com
David Allen	user5@example.com
Mark Duke	user6@example.com
Briana Wright	user7@example.com
John Bryan	user8@example.com
Jason Thompson	user9@example.com
Shawn Hill	user10@example.com

customers 3

3

```
select order_item_id, order_id, product_id, quantity, item_price, (quantity * item_price) as Total_Value
from order_items;
```

SQL File 8 | SQL File 8* | SQL File 12* | retail_data_DMLS

Limit to 1000 rows

```
27    order by price desc, name asc;
28
29 •  use project;
30
31 •  select o.order_id, o.customer_id, o.order_date, o.status, o.total_amount from orders o
32    left join customers c on o.customer_id = c.customer_id
33    where c.customer_id is null;
34
35 •  select name as Customer_Name, email as Customer_Email from customers;
36
37 •  select order_item_id, order_id, product_id, quantity, item_price, (quantity * item_price) as Total_Value
38    from order_items
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

order_item_id	order_id	product_id	quantity	item_price	Total_Value
1	1	19	2	4707.14	9414.28
2	2	6	3	177.40	532.20
3	3	9	3	985.19	2955.57
4	3	23	1	2208.99	2208.99
5	4	35	2	4734.89	9469.78
6	5	19	1	4707.14	4707.14
7	5	7	2	4897.36	9794.72
8	6	7	3	4897.36	14692.08
9	6	37	1	1429.45	1429.45
10	6	15	1	723.97	723.97

Result 4

4

select concat(name, '_', phone) as Customer_contact from customers;

SQL File 8 | SQL File 8* | SQL File 12* | retail_data_DMLS

Limit to 1000 rows

```
select o.order_id, o.customer_id, o.order_date, o.status, o.total_amount from orders o
left join customers c on o.customer_id = c.customer_id
where c.customer_id is null;

select name as Customer_Name, email as Customer_Email from customers;

select order_item_id, order_id, product_id, quantity, item_price, (quantity * item_price) as Total_Value
from order_items;

select concat(name, '_', phone) as Customer_contact from customers
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_contact
Thomas Owens_142-479-1945
Charles Grant_9153947511
Kaitlin Richards_2073473421
Christina Williams_586-605-5061x06
David Allen_(751)456-8289x1
Mark Duke_(144)957-2811
Briana Wright_223-833-9635
John Bryan_045.568.0798x27
Jason Thompson_1862659420
Shawn Hill_(268)113-3152x7

Result 5

5

select order_id, date(order_date) as Order_Date_only, status,total_amount from orders;

SQL File 8 | SQL File 8* | SQL File 12* | retail_data_DMLS

Limit to 1000 rows

```
left join customers c on o.customer_id = c.customer_id
where c.customer_id is null;

select name as Customer_Name, email as Customer_Email from customers;

select order_item_id, order_id, product_id, quantity, item_price, (quantity * item_price) as Total_Value
from order_items;

select concat(name, '_', phone) as Customer_contact from customers;

select order_id, date(order_date) as Order_Date_only, status,total_amount from orders
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

order_id	Order_Date_only	status	total_amount
1	2025-03-02	Delivered	9414.28
2	2024-10-09	Shipped	532.20
3	2025-05-08	Cancelled	5164.56
4	2024-09-19	Delivered	9469.78
5	2025-04-08	Pending	14501.86
6	2024-10-25	Cancelled	31050.17
7	2024-07-29	Shipped	3043.67
8	2024-07-30	Cancelled	32714.06
9	2025-06-10	Pending	24219.20
10	2025-02-16	Delivered	24342.52

Result 7

6

select product_id, name, category, price, stock_quantity from products
where stock_quantity = 0;

SQL File 8 | SQL File 8* | SQL File 12* | retail_data_DMLS

Limit to 1000 rows

```
select name as Customer_Name, email as Customer_Email from customers;

select order_item_id, order_id, product_id, quantity, item_price, (quantity * item_price) as Total_Value
from order_items;

select concat(name, '_', phone) as Customer_contact from customers;

select order_id, date(order_date) as Order_Date_only, status,total_amount from orders;

select product_id, name, category, price, stock_quantity from products
where stock_quantity = 0;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

product_id	name	category	price	stock_quantity
NULL	NULL	NULL	NULL	NULL

products 11

Level 3: Aggregations

1
select count(*) as Total_Orders from orders;

```
select order_item_id, order_id, product_id, quantity, item_price, (
from order_items;

select concat(name, '_', phone) as Customer_contact from customers;

select order_id, date(order_date) as Order_Date_only, status,total_

select product_id, name, category, price, stock_quantity from produ
where stock_quantity = 0;

select count(*) as Total_Orders from orders
```

Total_Orders
400

2
select sum(amount_paid) as Total_Revenue from payments;

```
select concat(name, '_', phone) as Customer_contact from customers;

select order_id, date(order_date) as Order_Date_only, status,total_am

select product_id, name, category, price, stock_quantity from product
where stock_quantity = 0;

select count(*) as Total_Orders from orders;

select sum(amount_paid) as Total_Revenue from payments
```

Total_Revenue
6960973.66

3
select avg(total_amount) as Average_Order_Value
from orders;

```
select order_id, date(order_date) as Order_Date_only, status,t

select product_id, name, category, price, stock_quantity from
where stock_quantity = 0;

select count(*) as Total_Orders from orders;

select sum(amount_paid) as Total_Revenue from payments;

select avg(total_amount) as Average_Order_Value
from orders
```

Average_Order_Value
17402.434150

4
select count(distinct customer_id) as Active_Customers
from orders;

```
where stock_quantity = 0;

select count(*) as Total_Orders from orders;

select sum(amount_paid) as Total_Revenue from payments;

select avg(total_amount) as Average_Order_Value
from orders;

select count(distinct customer_id) as Active_Customers
from orders;
```

Active_Customers
30

5
select c.customer_id, c.name, COUNT(o.order_id) AS total_orders
from customers c
join orders o ON c.customer_id = o.customer_id
GROUP BY c.customer_id, c.name
ORDER BY total_orders DESC;

```
select avg(total_amount) as Average_Order_Value
from orders;

select count(distinct customer_id) as Active_Customers
from orders;

select c.customer_id, c.name, COUNT(o.order_id) AS total_orders
from customers c
join orders o ON c.customer_id = o.customer_id
GROUP BY c.customer_id, c.name
ORDER BY total_orders DESC;
```

customer_id	name	total_orders
29	Kara Zavala	20
2	Charles Grant	17
3	Kaitlin Richards	17
17	Randy Mooney	17
20	Megan Lee	17
28	Joseph Stuart	17
6	Mark Duke	16
16	Amy Landry	16
30	Victoria Acevedo	16

6

```
select c.customer_id, c.name, SUM(p.amount_paid) AS total_spent
from customers c
join orders o ON c.customer_id = o.customer_id
join payments p ON o.order_id = p.order_id
GROUP BY c.customer_id, c.name
ORDER BY total_spent DESC;
```

SQL File 8 | SQL File 8* | SQL File 12* | retail_data_DMLS

Limit to 1000 rows

```
57 • select c.customer_id, c.name, COUNT(o.order_id) AS total_orders
58   from customers c
59   join orders o ON c.customer_id = o.customer_id
60   GROUP BY c.customer_id, c.name
61   ORDER BY total_orders DESC;
62
63 • select c.customer_id, c.name, SUM(p.amount_paid) AS total_spent
64   from customers c
65   join orders o ON c.customer_id = o.customer_id
66   join payments p ON o.order_id = p.order_id
67   GROUP BY c.customer_id, c.name
68   ORDER BY total_spent DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	name	total_spent
29	Kara Zavala	435408.89
24	Brandy Wright	379286.82
28	Joseph Stuart	362584.19
14	Deborah Arias	360324.18
25	Cindy Hart	335100.94
2	Charles Grant	284420.07
30	Victoria Acevedo	281841.38
16	Amy Landry	278469.73
5	David Allen	262504.19

Result 4

7

```
select p.category, SUM(oi.quantity) AS total_products_sold
from order_items oi
join products p ON oi.product_id = p.product_id
GROUP BY p.category
ORDER BY total_products_sold desc;
```

SQL File 8 | SQL File 8* | SQL File 12* | retail_data_DMLS

Limit to 1000 rows

```
63 • select c.customer_id, c.name, SUM(p.amount_paid) AS total_spent
64   from customers c
65   join orders o ON c.customer_id = o.customer_id
66   join payments p ON o.order_id = p.order_id
67   GROUP BY c.customer_id, c.name
68   ORDER BY total_spent DESC;
69
70 • select p.category, SUM(oi.quantity) AS total_products_sold
71   from order_items oi
72   join products p ON oi.product_id = p.product_id
73   GROUP BY p.category
74   ORDER BY total_products_sold desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	total_products_sold
Electronics	687
Clothing	559
Home	443
Toys	405
Books	350

8

```
select p.category, avg(oi.item_price) as avg_item_price
from order_items oi
join products p on oi.product_id = p.product_id
group by p.category;
```

SQL File 8 | SQL File 8* | SQL File 12* | retail_data_DMLS

Limit to 1000 rows

```
68   ORDER BY total_spent DESC;
69
70 • select p.category, SUM(oi.quantity) AS total_products_sold
71   from order_items oi
72   join products p ON oi.product_id = p.product_id
73   GROUP BY p.category
74   ORDER BY total_products_sold desc;
75
76 • select p.category, avg(oi.item_price) as avg_item_price
77   from order_items oi
78   join products p on oi.product_id = p.product_id
79   group by p.category;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	avg_item_price
Home	2330.711435
Clothing	3473.168292
Electronics	2733.925676
Toys	2407.628408
Books	3216.389412

9

```
select date(order_date) as order_day, COUNT(*) as orders_count
from orders
group by date(order_date)
order by order_day;
```

SQL File 8 | SQL File 8* | SQL File 12* | retail_data_DMLS

Limit to 1000 rows

```
GROUP BY p.category
ORDER BY total_products_sold desc;

select p.category, avg(oi.item_price) as avg_item_price
from order_items oi
join products p on oi.product_id = p.product_id
group by p.category;

select date(order_date) as order_day, COUNT(*) as orders_count
from orders
group by date(order_date)
order by order_day;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

order_day	orders_count
2024-06-15	1
2024-06-18	1
2024-06-19	3
2024-06-20	2
2024-06-21	2
2024-06-23	2
2024-06-24	1
2024-06-25	1
2024-06-30	2

Result 7

10

```
select method, SUM(amount_paid) as total_collected
from payments
group by method
order by total_collected desc;
```

SQL File 8 | SQL File 8* | SQL File 12* | retail_data_DMLS

Limit to 1000 rows

```
join products p on oi.product_id = p.product_id
group by p.category;

select date(order_date) as order_day, COUNT(*) as orders_count
from orders
group by date(order_date)
order by order_day;

select method, SUM(amount_paid) as total_collected
from payments
group by method
order by total_collected desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

method	total_collected
Debit Card	1930577.88
Credit Card	1754603.10
Net Banking	1658383.90
UPI	1617408.78

Level 4: Multi-Table Queries (JOINS)

1

```
select o.order_id, o.order_date, o.status, o.total_amount,
c.name AS customer_name
from orders o
inner join customers c on o.customer_id = c.customer_id;
```

SQL File 8 | SQL File 8* | SQL File 12* | retail_data_DMLS

Limit to 1000 rows

```
83  group by date(order_date)
84  order by order_day;
85
86  select method, SUM(amount_paid) as total_collected
87  from payments
88  group by method
89  order by total_collected desc;
90
91  select o.order_id, o.order_date, o.status, o.total_amount,
92  c.name AS customer_name
93  from orders o
94  inner join customers c on o.customer_id = c.customer_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

order_id	order_date	status	total_amount	customer_name
14	2024-09-24 21:21:38	Shipped	15803.34	Thomas Owens
17	2024-08-19 21:17:57	Pending	11173.77	Thomas Owens
61	2024-12-26 23:21:58	Shipped	13053.00	Thomas Owens
76	2025-01-14 22:59:54	Pending	23506.81	Thomas Owens
92	2024-09-26 11:33:58	Shipped	834.75	Thomas Owens
109	2025-03-17 04:56:18	Shipped	12190.14	Thomas Owens
127	2025-06-10 18:01:32	Pending	20160.64	Thomas Owens
135	2025-03-11 22:58:23	Pending	8891.79	Thomas Owens
144	2024-09-19 09:18:22	Shipped	12093.54	Thomas Owens

Result 10

2

```
select distinct p.product_id, p.name as product_name,
p.category, p.price
from products p
inner join order_items oi on p.product_id = oi.product_id;
```

SQL File 8 | SQL File 8* | SQL File 12* | retail_data_DMLS

Limit to 1000 rows

```
87  from payments
88  group by method
89  order by total_collected desc;
90
91  select o.order_id, o.order_date, o.status, o.total_amount,
92  c.name AS customer_name
93  from orders o
94  inner join customers c on o.customer_id = c.customer_id;
95
96  select distinct p.product_id, p.name as product_name, p.category, p.price
97  from products p
98  inner join order_items oi on p.product_id = oi.product_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product_id	product_name	category	price
1	Plant No	Home	639.43
2	Population Social	Clothing	4813.68
3	Available Answer	Electronics	2529.51
4	Any Question	Clothing	4759.28
5	Natural Network	Toys	4722.66
6	If Whatever	Electronics	177.40
7	Response Indeed	Clothing	4897.36
8	Every Amount	Home	4173.60
9	Common Study	Toys	985.19

Result 11

Output

3

```
select o.order_id, o.total_amount, p.method as
payment_method, p.amount_paid
from orders o
inner join payments p on o.order_id = p.order_id;
```

SQL File 8 | SQL File 8* | SQL File 12* | retail_data_DMLS

Limit to 1000 rows

```
91  select o.order_id, o.order_date, o.status, o.total_amount,
92  c.name AS customer_name
93  from orders o
94  inner join customers c on o.customer_id = c.customer_id;
95
96  select distinct p.product_id, p.name as product_name, p.category, p.price
97  from products p
98  inner join order_items oi on p.product_id = oi.product_id;
99
100 select o.order_id, o.total_amount, p.method as payment_method, p.amount_paid
101 from orders o
102 inner join payments p on o.order_id = p.order_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

order_id	total_amount	payment_method	amount_paid
1	9414.28	Credit Card	9414.28
2	532.20	Net Banking	532.20
3	5164.56	Credit Card	5164.56
4	9469.78	UPI	9469.78
5	14501.86	UPI	14501.86
6	31050.17	UPI	31050.17
7	3043.67	UPI	3043.67
8	32714.06	Net Banking	32714.06
9	24219.20	Net Banking	24219.20

4

```
select c.customer_id, c.name AS customer_name, o.order_id, o.order_date,
o.status
from customers c
left join orders o on c.customer_id = o.customer_id
order by c.customer_id;
```

SQL File 8 | SQL File 8* | SQL File 12* | retail_data_DMLS

Limit to 1000 rows

```
96  select distinct p.product_id, p.name as product_name, p.category, p.price
97  from products p
98  inner join order_items oi on p.product_id = oi.product_id;
99
100 select o.order_id, o.total_amount, p.method as payment_method, p.amount_paid
101 from orders o
102 inner join payments p on o.order_id = p.order_id;
103
104 select c.customer_id, c.name AS customer_name, o.order_id, o.order_date, o.status
105 from customers c
106 left join orders o on c.customer_id = o.customer_id
107 order by c.customer_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	customer_name	order_id	order_date	status
1	Thomas Owens	14	2024-09-24 21:21:38	Shipped
1	Thomas Owens	17	2024-08-19 21:17:57	Pending
1	Thomas Owens	61	2024-12-26 23:21:58	Shipped
1	Thomas Owens	76	2025-01-14 22:59:54	Pending
1	Thomas Owens	92	2024-09-26 11:33:58	Shipped
1	Thomas Owens	109	2025-03-17 04:56:18	Shipped
1	Thomas Owens	127	2025-06-10 18:01:32	Pending
1	Thomas Owens	135	2025-03-11 22:58:23	Pending
1	Thomas Owens	144	2024-09-19 09:18:22	Shipped

5

```
select p.product_id, p.name AS product_name,
ifnull(SUM(oi.quantity), 0) AS total_quantity_sold
from products p
left join order_items oi on p.product_id = oi.product_id
group by p.product_id, p.name
order by total_quantity_sold desc;
```

6

```
select p.payment_id, p.order_id, p.amount_paid, p.method,
o.order_date, o.status
from payments p
right join orders o on p.order_id = o.order_id;
```

SQL File 8 | SQL File 8* | SQL File 12* | retail_data_DMLS

Limit to 1000 rows

```
select c.customer_id, c.name AS customer_name, o.order_id, o
from customers c
left join orders o on c.customer_id = o.customer_id
order by c.customer_id;

select p.product_id, p.name AS product_name,
ifnull(SUM(oi.quantity), 0) AS total_quantity_sold
from products p
left join order_items oi on p.product_id = oi.product_id
group by p.product_id, p.name
order by total_quantity_sold desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product_id	product_name	total_quantity_sold
24	Assume Serve	83
46	Age Treatment	67
18	Some Them	63
47	Southern Thing	62
8	Every Amount	61
22	Move Small	60
20	Real Source	60
37	Between Up	58
17	Everything Plant	58

Result 16

SQL File 8 | SQL File 8* | SQL File 12* | retail_data_DMLS

Limit to 1000 rows

```
select p.product_id, p.name AS product_name,
ifnull(SUM(oi.quantity), 0) AS total_quantity_sold
from products p
left join order_items oi on p.product_id = oi.product_id
group by p.product_id, p.name
order by total_quantity_sold desc;

select p.payment_id, p.order_id, p.amount_paid, p.method,
o.order_date, o.status
from payments p
right join orders o on p.order_id = o.order_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

payment_id	order_id	amount_paid	method	order_date	status
1	1	9414.28	Credit Card	2025-03-02 07:20:11	Delivered
2	2	532.20	Net Banking	2024-10-09 18:08:21	Shipped
3	3	5164.56	Credit Card	2025-05-08 00:08:27	Cancelled
4	4	9469.78	UPI	2024-09-19 22:16:13	Delivered
5	5	14501.86	UPI	2025-04-08 18:02:06	Pending
6	6	31050.17	UPI	2024-10-25 07:33:59	Cancelled
7	7	3043.67	UPI	2024-07-29 11:58:47	Shipped
8	8	32714.06	Net Banking	2024-07-30 22:49:49	Cancelled
9	9	24219.20	Net Banking	2025-06-10 17:00:25	Pending

Result 17

7

```
SELECT c.name AS customer_name, c.email,
o.order_id, o.order_date, o.total_amount, o.status,
p.payment_id, p.amount_paid, p.method AS payment_method, p.payment_date
FROM customers c
INNER JOIN orders o ON c.customer_id = o.customer_id
INNER JOIN payments p ON o.order_id = p.order_id
ORDER BY o.order_date DESC;
```

SQL File 8 | SQL File 8* | SQL File 12* | retail_data_DMLS

Limit to 1000 rows

```
select p.payment_id, p.order_id, p.amount_paid, p.method,
o.order_date, o.status
from payments p
right join orders o on p.order_id = o.order_id;

SELECT c.name AS customer_name, c.email,
o.order_id, o.order_date, o.total_amount, o.status,
p.payment_id, p.amount_paid, p.method AS payment_method, p.payment_date
FROM customers c
INNER JOIN orders o ON c.customer_id = o.customer_id
INNER JOIN payments p ON o.order_id = p.order_id
ORDER BY o.order_date DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_name	email	order_id	order_date	total_amount	status	payment_id	amount_paid	payment_method	payment_date
Shawn Hill	user10@example.com	62	2025-06-13 23:21:01	35723.17	Cancelled	62	35723.17	Credit Card	2025-06-13 23:45:01
Amanda Bright	user19@example.com	173	2025-06-13 10:58:21	6875.62	Shipped	173	6875.62	UPI	2025-06-13 11:43:21
Jason Thompson	user9@example.com	356	2025-06-11 13:30:19	11584.94	Pending	356	11584.94	Net Banking	2025-06-11 14:21:19
Austin Flores	user15@example.com	286	2025-06-11 08:36:00	26952.09	Pending	286	26952.09	UPI	2025-06-11 08:47:00
Deborah Arias	user14@example.com	308	2025-06-11 03:03:58	36411.35	Cancelled	308	36411.35	Debit Card	2025-06-11 03:45:58
Thomas Owens	user1@example.com	127	2025-06-10 18:01:32	20160.64	Pending	127	20160.64	Debit Card	2025-06-10 18:57:32
Mark Duke	user6@example.com	9	2025-06-10 17:00:25	24219.20	Pending	9	24219.20	Net Banking	2025-06-10 17:36:25
Cindy Hart	user25@example.com	313	2025-06-09 19:00:34	24245.67	Cancelled	313	24245.67	Credit Card	2025-06-09 19:41:34
Kaitlin Richards	user3@example.com	134	2025-06-09 13:59:27	15483.53	Pending	134	15483.53	Net Banking	2025-06-09 14:20:27

Result 18

Level 5: Subqueries (Inner Queries)

1

```
select product_id, name, category, price
from products
where price > (select avg(price) from products);
```

SQL File 8 | SQL File 8* | SQL File 12* | retail_data_DMLS

```
SELECT c.name AS customer_name, c.email,
o.order_id, o.order_date, o.total_amount, o.status,
p.payment_id, p.amount_paid, p.method AS payment_method
FROM customers c
INNER JOIN orders o ON c.customer_id = o.customer_id
INNER JOIN payments p ON o.order_id = p.order_id
ORDER BY o.order_date DESC;

select product_id, name, category, price
from products
where price > (select avg(price) from products);
```

Result Grid

product_id	name	category	price
2	Population Social	Clothing	4813.68
4	Any Question	Clothing	4759.28
5	Natural Network	Toys	4722.66
7	Response Indeed	Clothing	4897.36
8	Every Amount	Home	4173.60
10	Development System	Electronics	4801.78
12	Action Ask	Electronics	4017.01
18	Some Them	Toys	3673.86
19	Build High	Clothing	4707.14

products 19

2

```
select customer_id, name, email
from customers
where customer_id in (select distinct customer_id from
```

SQL File 8 | SQL File 8* | SQL File 12* | retail_data_DMLS

```
FROM customers c
INNER JOIN orders o ON c.customer_id = o.customer_id
INNER JOIN payments p ON o.order_id = p.order_id
ORDER BY o.order_date DESC;

select product_id, name, category, price
from products
where price > (select avg(price) from products);

select customer_id, name, email
from customers
where customer_id in (select distinct customer_id from orders);
```

Result Grid

customer_id	name	email
1	Thomas Owens	user1@example.com
2	Charles Grant	user2@example.com
3	Kaitlin Richards	user3@example.com
4	Christina Williams	user4@example.com
5	David Allen	user5@example.com
6	Mark Duke	user6@example.com
7	Briana Wright	user7@example.com
8	John Bryan	user8@example.com
9	Jason Thompson	user9@example.com

customers 20

3

```
select o.order_id, o.customer_id, o.total_amount
from orders o
where total_amount > (
    select avg(total_amount)
    from orders
    where customer_id = o.customer_id);
```

SQL File 8 | SQL File 8* | SQL File 12* | retail_data_DMLS

```
where price > (select avg(price) from products);

select customer_id, name, email
from customers
where customer_id in (select distinct customer_id

select o.order_id, o.customer_id, o.total_amount
from orders o
where total_amount > (
    select avg(total_amount)
    from orders
    where customer_id = o.customer_id);
```

Result Grid

order_id	customer_id	total_amount
6	29	31050.17
8	19	32714.06
9	6	24219.20
10	28	24342.52
14	1	15803.34
16	5	22856.73
18	13	32001.24
21	22	25364.11
22	20	21281.44

orders 21

4

```
select customer_id, name, email
from customers
where customer_id not in (select distinct customer_id from orders);
```

SQL File 8 | SQL File 8* | SQL File 12* | retail_data_DMLS

```
where customer_id in (select distinct customer_id from orders);

select o.order_id, o.customer_id, o.total_amount
from orders o
where total_amount > (
    select avg(total_amount)
    from orders
    where customer_id = o.customer_id);

select customer_id, name, email
from customers
where customer_id not in (select distinct customer_id from orders);
```

Result Grid

customer_id	name	email
NULL	NULL	NULL

5

```
select product_id, name, category, price
from products
where product_id not in (select distinct product_id from order_items);
```

SQL File 8 | SQL File 8* | SQL File 12* | retail_data_DMLS

```
where total_amount > (
    select avg(total_amount)
    from orders
    where customer_id = o.customer_id);

select customer_id, name, email
from customers
```

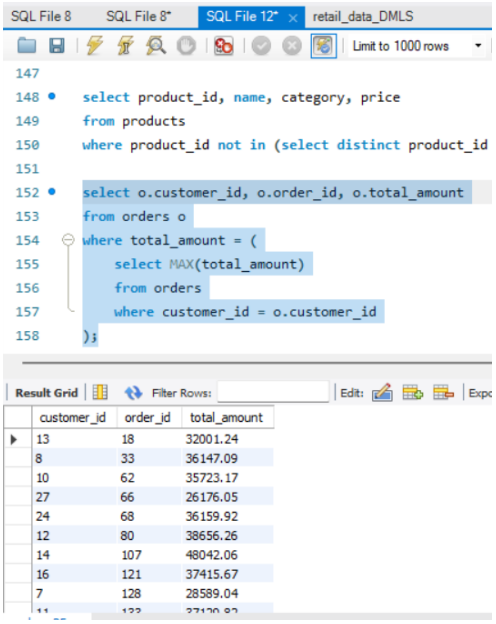
5

```
select product_id, name, category, price
from products
where product_id not in (select distinct product_id from order_items);
```

product_id	name	category	price
NULL	NULL	NULL	NULL

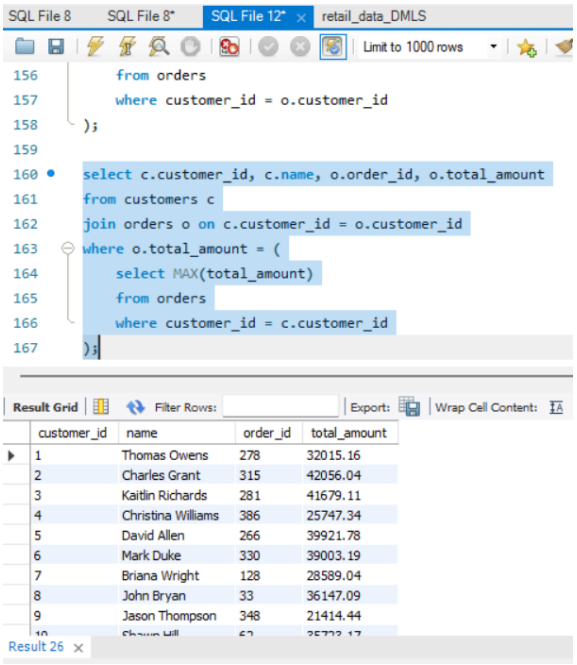
6

```
select o.customer_id, o.order_id, o.total_amount
from orders o
where total_amount = (
   select MAX(total_amount)
   from orders
   where customer_id = o.customer_id
);
```



7

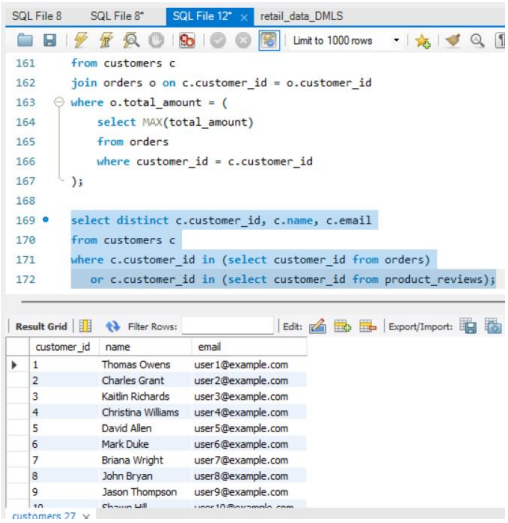
```
select c.customer_id, c.name, o.order_id, o.total_amount
from customers c
join orders o on c.customer_id = o.customer_id
where o.total_amount = (
   select MAX(total_amount)
   from orders
   where customer_id = c.customer_id
);
```

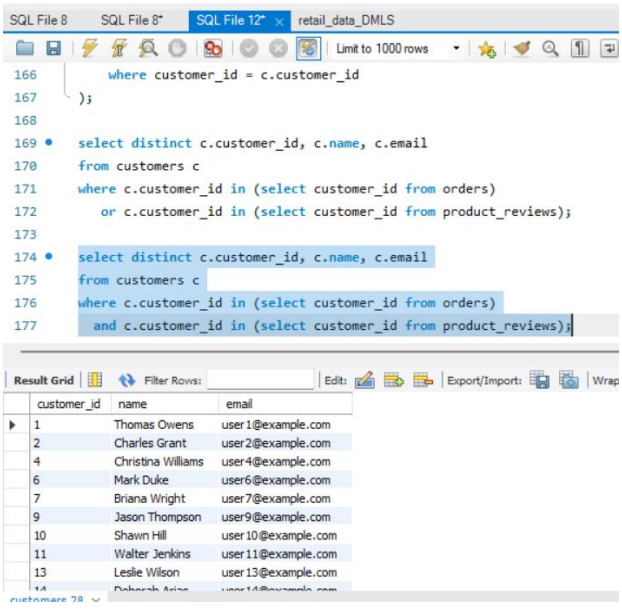


Level 6: Set Operations

1

```
select distinct c.customer_id, c.name, c.email
from customers c
where c.customer_id in (select customer_id from orders)
  or c.customer_id in (select customer_id from product_reviews);
```



2

```
select distinct c.customer_id, c.name, c.email
from customers c
where c.customer_id in (select customer_id from orders)
  and c.customer_id in (select customer_id from product_reviews);
```